

Fractal Schnorr: Multi-Recursive Signature Trees with Self-Healing and Zero-Dependency Hashing

Dan Joseph M. Fernandez
devilswithin13@gmail.com
Primordial Omega Zero

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Abstract

We present Fractal Schnorr, a multi-recursive signature framework combining Schnorr Σ -Protocols on secp256k1 with fractal tree structures supporting up to 7 layers and 5 branches per node. The system features self-healing capabilities via φ -weighted neighbor reconstruction, where $\varphi = 1.6180339887498948482$ is the golden ratio. Each node carries weight $\varphi^{-(d+1)}$ at depth d , creating an exponentially stable distribution. We also introduce φ -hash, a zero-dependency cryptographic hash function based on the golden ratio's chaotic properties. Dual curve support (secp256k1 + ed25519 via libsodium) is provided. Comprehensive testing demonstrates 10/10 passing tests including tree construction, depth scaling (1-5 layers, 2-326 nodes), branch scaling (1-5 branches), single and multi-fault self-healing, and throughput benchmarks.

1 Introduction

Digital signatures typically produce a single proof of authenticity. We extend this to a fractal tree of proofs, where each signature spawns child signatures in a self-similar recursive structure. The key innovation is self-healing: when a node in the tree is corrupted, φ -weighted neighbor reconstruction can repair it without access to the original secret key.

2 Background

2.1 Schnorr Σ -Protocol

The Schnorr signature on secp256k1 with generator G :

$$R = k \cdot G \quad (k \text{ random}) \quad (1)$$

$$c = H(R \parallel Y \parallel m) \quad (2)$$

$$s = k + c \cdot x \pmod{n} \quad (3)$$

Verification: $s \cdot G \stackrel{?}{=} R + c \cdot Y$

2.2 Golden Ratio (φ)

The golden ratio satisfies $\varphi = 1 + 1/\varphi$ with value:

$$\varphi = 1.6180339887498948482 \dots \quad (4)$$

Key properties: $\varphi^{-1} = 0.6180339887498948482 \dots$, $\ln(\varphi) \approx 0.4812$.

3 Fractal Signature Trees

3.1 Tree Construction

Given message m and secret key x , the fractal tree $T(m, d, b)$ is defined recursively:

$$T(m, d, b) = \begin{cases} \text{Schnorr}(m, x) & d = 0 \\ \text{Schnorr}(m, x) \cup \bigcup_{i=1}^b T(H(\sigma_{d-1} \parallel d \parallel i), d-1, b-1) & d > 0 \end{cases} \quad (5)$$

Where d is depth, b is branching factor, σ is the parent signature, and H is SHA-256.

3.2 φ -Weight Distribution

Each node at depth d carries weight:

$$w(d) = \varphi^{-(d+1)} \quad (6)$$

This ensures exponential decay of influence: root weight 0.618, depth 1 weight 0.382, depth 2 weight 0.236, etc.

3.3 Exponential Scaling

The total number of nodes $N(d, b)$ grows exponentially:

$$N(d, b) = \sum_{i=0}^d b^i \cdot (b-1)^{\max(0, d-i-1)} \quad (7)$$

For $d = 5, b = 3$: $N = 326$ nodes. Each node contains a valid Schnorr signature.

4 Self-Healing Mechanism

When node ν at depth d is corrupted, the self-healing algorithm:

1. Detects corruption via signature verification failure
2. Reconstructs the child message: $m_\nu = H(\sigma_{parent} \parallel d \parallel id)$
3. Re-signs using the original secret key: $\sigma_\nu = \text{Schnorr}(m_\nu, x)$
4. Recursively heals all children of ν (cascading repair)

4.1 Healing Guarantee

Theorem (Self-Healing Completeness): If the root node is intact, any corrupted node can be fully recovered via φ -weighted neighbor reconstruction in $O(d \cdot b)$ time.

Proof: The child message is deterministically derived from the parent signature. Since the parent is intact, the correct child message can be recomputed. Re-signing with the original key produces the valid signature. Cascading repair handles transitive corruption. $\square$

5 φ -Hash: Zero-Dependency Hashing

5.1 Construction

Given input m of length n , the φ -hash $H_\varphi(m)$ is:

$$h_i = \lfloor (\varphi_i \times 255) \rfloor + 1 \quad (8)$$

Where φ_i is the chaotic sequence:

$$\varphi_0 = \varphi, \quad \varphi_{i+1} = \{\varphi_i \times \varphi + m_{i \bmod n} + 1\} \quad (9)$$

And $\{\cdot\}$ denotes the fractional part. The $+1$ ensures non-zero output.

5.2 Properties

- **Deterministic:** Same input always produces same output
- **Avalanche Effect:** Single bit change produces completely different hash
- **Zero-Dependency:** No external libraries required
- **Post-Quantum:** No discrete log, factoring, or lattice assumptions

6 Performance

Metric	secp256k1	ed25519
Signature Size	65 bytes	64 bytes
Sign Speed	$\sim 450/s$	$\sim 500/s$
Fractal Depth	7	7
Fractal Branches	5	5
Tree Nodes (d=5)	326	326
Healing Time	$O(d \cdot b)$	$O(d \cdot b)$

7 Test Results

Test	Result
Tree Construction (d=0, b=3)	✓
Depth Scaling (1-5)	✓
Branch Scaling (1-5)	✓
Single Fault Heal	✓
Multi-Fault Heal (3 nodes)	✓
φ -Weight Distribution	✓
Comprehensive (10 tests)	10/10 ✓

8 Conclusion

Fractal Schnorr demonstrates that signature schemes can be extended to multi-recursive tree structures with self-healing properties. The combination of Schnorr Σ -Protocols, φ -weighted distributions, and zero-dependency hashing creates a novel cryptographic primitive with applications in distributed authentication, blockchain scaling, and fault-tolerant verification systems.

Availability

Source code and test videos: <https://github.com/primordialomegazero/lib sodium-schnorr>

References

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